

Building a Unit Conversion Application in App Inventor 2

If you haven't joined the Android app developer bandwagon yet, then welcome to App Inventor 2. If you are a first timer, we suggest you read our previous articles (which have been featured here for around a year now). If you've already got a few apps in your armoury, then good luck and carry on.



For the past year, we have embarked on a journey of developing apps using App Inventor. During this period, we developed multiple applications and I hope you have also experimented with things on your own. If your applications have been published in the Google Play store, I would like to check out your work; so please share your app download links via email.

In App Inventor, we have mastered the components available for use in the palette. Playing around with the designer and block editor is so much fun.

A unit conversion app or software is quite a necessity in our daily lives. There are multiple instances when we need to convert some unit to another and vice versa. The application comes handy for faster calculations. So let's develop a simple, yet fun filled Android utility application. You have probably practised unit conversion in mathematical problems in school. In this article, we will revise your knowledge of units.

The theme of the application

The theme is pretty simple -- we need to design and program the application so that it can convert one unit to another. Rather than doing single unit conversions, let's give users options to pick from a range of alternatives.

GUI requirements

For every application, we have a graphical user interface or GUI, which helps users to interact with the on-screen components. How each component responds to user actions is defined in the block editor section.

GUI requirements for Screen1

- Labels:** The labels are the static text components used to display some headings or markings on the screen.
- Button:** A button will let you trigger the event and is a very essential component.
- Horizontal arrangement:** These are special components which keep all child components aligned. Horizontal components keep all

the child components horizontally aligned.

- Notifier:** This is used to display some instructions or control your existing components. You will see its functionality in more detail as we implement it in our app.
- List picker:** This is the GUI component which presents a list to users, who have to choose from the displayed options.
- Input box:** These dedicated user input boxes allow the user to type the input. During the designing stage, you can set the type of accepted input, e.g., numerical only, multi-line, etc.

Table 1 lists the components that we will need for this application. Let us drag them to the designer from the left hand side palette.

- Drag and drop the components mentioned in the table above the viewer.
- Visible components can be seen by you, while the non-visible components will be located beneath